

Arush Bisht

650-730-3922 | arush.bijay.bisht@gmail.com | Foster City, CA | Class of 2027

EDUCATION

University of California - San Diego

La Jolla, CA

B.S. Mechanical Engineering (GPA: 3.8)

Expected 2027

- **Relevant Coursework:** Linear Control Theory, Computational Methods, Solid & Fluid Mechanics, Heat Transfer, Autonomous Vehicle Lab

EXPERIENCE

Students for the Exploration and Development of Space (SEDS)

Oct. 2024 – Present

Guidance, Navigation, and Control (GNC) Lead

La Jolla, CA

- Direct a 6-member subteam in the development of a lander testbed, managing cross-functional integration between mechanical, software, and electrical systems.
- Engineered a 324-part rocket CAD assembly, optimizing for **DFM** and 1.5 safety factor under 1000 lbf loads.
- Integrated **CAN-to-I2C** protocols between a Jetson Nano, flight computer, and four high-torque actuators.
- Designed a full rocket dynamics **LQR controller** for 6-DOF trajectory tracking and attitude stabilization, validating performance across the flight envelope with **Monte Carlo simulations** for robustness analysis.

Multi-Agent Robotics (MURO) Lab

Sept. 2025 – Present

Undergraduate Researcher

La Jolla, CA

- Developed a **Linear MPC** trajectory tracker for a multi-agent system; formulated a quadratic programming (QP) problem that improved cross-track error by 20% over standard PID.
- Implemented **feedback-linearization** and path-planning algorithms in **ROS2** for cargo-tethered **Linux** robot swarms, validating control performance in Gazebo simulations.
- Refactored lab-wide ROS2/Docker software stack, reducing sim-to-hardware integration time by **50%**.

Thermal Energy and Material Physics (TEMP) Lab

Feb. 2025 – June 2025

Research Intern

La Jolla, CA

- Configured DAQ (Data Acquisition) instrumentation for analyzing heat flux and cooling efficiency in MATLAB.
- Optimized fabrication of Peltier semiconductor cooling devices using photolithography and thin-film deposition techniques.

PROJECTS

Multiplicative Extended Kalman Filter (MEKF) | C++, Python, Eigen, pybind11

Jan. 2025 – Present

- Developed a 6-DOF state estimation filter for a lander testbed using a laser altimeter, IMU, and GPS sensor fusion.
- Designed a custom **Allan Variance** pipeline to characterize stochastic noise and random walk in MEMS inertial sensors, significantly improving filter convergence.
- Engineered a high-performance C++ core with a **pybind11** interface, implementing a cross-language validation suite in **pytest** to ensure numerical parity for real-time deployment.

Engine Throttle Controller | MATLAB, Simulink, Hardware-in-the-loop

May 2024 – Present

- Designed a **dual-loop (cascaded) PID controller** to regulate Mixture Ratio (MR) and Chamber Pressure.
- Characterized valve dynamics using MATLAB **System Identification**, achieving 92% model accuracy.
- Performed **Hardware-in-the-Loop (HITL)** and water flow testing to obtain valve coefficients (C_v), achieving <500ms rise time and <5% overshoot across the full throttle range.

Rocket Drone Platform | C++, OpenCV, Fusion 360, Mechatronics

March 2025 – Present

- Engineered a 3D printed spring-loaded linkage mechanism for drone release; minimized vibrational fatigue
- Developed a **computer vision autonomous landing system** using OpenCV; achieved <0.2s frame latency for pad detection.
- Programmed a C++ mission state machine on a Raspberry Pi/Pixhawk stack, managing autonomous transitions between rocket ascent, propeller deployment, and landing modes.

TECHNICAL SKILLS

Controls/Robotics: ROS2, MPC, Kalman Filter, LQR, PID, Gazebo, System Identification.

Software/Math: MATLAB, Simulink, Python (NumPy, OpenCV, Pybind), C++ (Eigen), Docker, Git.

Mechanical/CAD: SolidWorks, Fusion 360, Ansys FEA, GD&T, DFM, Structural Analysis.

Fabrication: 3D Printing, CNC Machining, Waterjet Cutting, Soldering, Rapid Prototyping, Strain-Gauge DAQ.